

1. The most important role of a surfactant is:
 - a. To increase the surface tension at an oil-water interface
 - b. To provide stability of oil drops in water ← **CORRECT**
 - c. To provide a mechanism to transport fluid from one droplet to another

2. Which of the following is most effective at thickening a sauce and can be added at the lowest concentration?
 - a. Amylose (linear starch) ← **CORRECT**
 - b. Amylopectin (branched starch)
 - c. Sucrose (table sugar)

3. If the volume fraction of a foam is below the critical volume fraction, Φ_c , what is the elastic modulus?
0 Pa

4. A Ramos Gin Fizz is a cocktail made with a foam that is strong enough to support a metal straw. If you wanted a stiffer foam would you want to:
 - create smaller bubbles -> **correct**
 - create bigger bubbles
 - incorporate more air, i.e. more shaking -> **correct**
 - less air, i.e. gentle stirring

5. Grant Achatz is developing a cocktail for his restaurant, using a new thickener. He shares the cocktail with a physicist friend who knows how to measure the properties of liquids. His friend finds that the elastic modulus of the new cocktail is four times that of an older cocktail that was made using xanthan gum, and that the relaxation time for the new cocktail is half that of the old one.

a) Given that the viscosity of the old cocktail is 100 mPa·s, what is the viscosity of Grant's new cocktail? [5 pts]

Grant's friend found that:

$$E_{new} = 4E_{old} \text{ and } \tau_{new} = \frac{1}{2}\tau_{old}$$

Since $\eta = E \times \tau$

We can plug in the values above and solve for the viscosity:

$$\eta_{new} = E_{new} \times \tau_{new}$$

$$\eta_{new} = 4E_{old} \times \frac{1}{2} \tau_{old} = 2E_{old} \times \tau_{old} = 2\eta_{old} = 2 \times 100 \text{ mPa} = 200 \text{ mPa} \cdot \text{s}$$

□

The new viscosity is **200mPa·s**

6. Lactonnaise has 700 mL of oil dispersed in 300 mL of water. The radius of the droplets is roughly 10 microns and the surface tension of the oil water is about 20 N/m. What is the elasticity of the emulsion?

$$E = \frac{\sigma}{R} (\phi - \phi_c)$$

$$E = \frac{(20 \text{ N/m})}{(10^{-5} \text{ m})} (0.70 - 0.64)$$

$$E = 1.2 \times 10^5 \text{ Pa}$$

7. How long does it take for the microbial population inside a block of cheese take to reach 100 million, assuming the initial population was 1000 microbes and the doubling time is 40 minutes?

$$N = N_0 e^{kt}$$

$$k = \ln(2) / \tau = \ln(2) / (40 \text{ min})$$

$$k = 1.7 \times 10^{-2} \text{ min}^{-1}$$

$$t = \ln(N / N_0) / k$$

$$t = \ln(10^5) / (1.7 \times 10^{-2} \text{ min})$$

$$t = 664 \text{ min} \approx 10 \text{ hours}$$

8. For a vegetarian Thanksgiving, you cook a cylinder of Tofurkey that is 8 cm tall and 30 cm in diameter. The center of the fake meat needs to reach a temperature of 70 °C, starting from an initial refrigerator temperature of 4 °C. Assume the Tofurkey is mainly water. The oven is set to 175 °C.

$$t_{cook} = \frac{L^2}{\pi D_{heat}} \ln \left(\frac{T_{ext} - T_0}{T_{ext} - T_{target}} \right)$$

$$t_{cook} = \frac{(4\text{cm})^2}{\pi (1.4 \times 10^{-3} \text{cm}^2/\text{s})} \ln \left(\frac{175^\circ\text{C} - 4^\circ\text{C}}{175^\circ\text{C} - 70^\circ\text{C}} \right)$$

$$t_{cook} = 1775 \text{sec} \approx 30 \text{min}$$

9. The protein chains in a piece of soft cheese rearrange with a time scale of 10 hours. The elasticity of the cheese is 10kPa, but the cheese slowly flows over the course of several days. What is the viscosity of the cheese?

$$\eta = E \times \tau$$

$$\eta = (10 \text{kPa}) (10 \text{h} \times 3600 \text{s/h})$$

$$\eta = 3.6 \times 10^5 \text{ kPa} \cdot \text{s}$$

$$\eta = 3.6 \times 10^8 \text{ Pa} \cdot \text{s}$$

10. After you cut (or bite) into an apple, the surface will often turn brown, due to the action of an enzyme called polyphenol oxidase. To prevent browning, chefs often slice apples and immediately place them into a bath of water with a little lemon juice squeezed into it. Propose an explanation for why this method works.

Since the browning is due to an enzyme, the acidity probably denatures the enzyme or otherwise halts its activity

11. Anna really missed having Australian-style yogurt, so she smuggled some back. When the tub was almost finished, she took 20mL of the yogurt out and put it in 1L of fresh milk in an attempt to make more yogurt.

11 b. If the initial concentration of yogurt-making bacteria was 1000 bacteria/mL and the time constant for bacterial growth is 30mins, how long will it take before her 1L of milk has the same concentration of bacteria as the initial 20mL of yogurt?

$$N_0 = 1000 \times 20 = 20\,000 \text{ bacteria}$$

$$N_f = 1000 \times 1000 = 1\,000\,000 \text{ bacteria}$$

$$N_f = N_0 \exp(kt) \quad \rightarrow \quad N_f = N_0 \exp((\ln 2) t / \tau) \quad \rightarrow \quad t = (\tau \times \ln(N_f/N_0)) / \ln 2$$

Plugging in numbers:

$$t = ((30 \text{min} \times 60 \text{sec}) \times \ln(50)) / \ln 2 = 2.8 \text{ hours}$$

12. You would like to make a novel emulsion of hazelnut oil in coffee. You have achieved an emulsion, but for culinary reasons, you would like to maximize the elasticity. On your shelf you have two surfactants that you recently studied for this very purpose. For Surfactant A, you found that you could decrease your oil droplet size by 75% compared to your original test, but that surface tension was reduced by 50%. Surfactant B only reduced surface tension by 25%, and droplet size decreased by 50%. If you emulsify each trial just until it becomes stiff, which Surfactant should result in the highest elasticity, and more importantly why? To do this, you need to find how much each surfactant increases the original elasticity. What is one other thing you could do to increase the elasticity?

The equation for elasticity of an emulsion is $E = \frac{\sigma}{R}(\phi - \phi_c)$. For Surfactant A, the droplet size, R , is equal to $\frac{1}{4}$ of its original value, while the surface tension is equal to $\frac{1}{2}$ of its original value. The elasticity becomes $E_1 = \frac{(1/2)\sigma}{(1/4)R}(\phi - \phi_c) = \frac{2\sigma}{R}(\phi - \phi_c) = 2E$.

For Surfactant B, the droplet size decreases to $\frac{1}{2}$ of its value and the surface tension decreases to $\frac{3}{4}$ of its value. $E_2 = \frac{(3/4)\sigma}{(1/2)R}(\phi - \phi_c) = \frac{3\sigma}{2R}(\phi - \phi_c) = \frac{3}{2}E$.

Therefore, Surfactant A is the best choice.

To increase the elasticity more, we should increase the volume fraction above the critical volume fraction as much as possible.